

Dear Editor,

We submit the manuscript “Auditable repeated-service contracts for distribution-connected batteries: separating feeder limits from persistent recovery state” for consideration in *Applied Energy*.

The manuscript addresses a practical gap in flexibility assessment: a battery offer that is feasible for one activation is not necessarily feasible after a partially completed recovery. We define a sequence-level service-recovery contract and audit every interval through an isolated three-phase OpenDSS snapshot. The experiment includes a pre-registered full-rearm negative control. This control is essential because it distinguishes a genuine persistent SOC effect from a changing feeder scenario or a solver-state artifact. On 214 training, 71 calibration, and 80 held-out Ausgrid customer-days, a frozen 140 kW offer reaches 96.25% held-out event success. The full-rearm capacity remains at 140 kW through six events, while the reserve-limited contract falls to 40 kW by the sixth event.

The work is aligned with the journal’s interest in energy-system flexibility, distributed storage, and distribution-network operation. It complements operating-envelope, recovery-aware bidding, and AC flexibility studies by providing a reproducible acceptance interface for repeated services. The package contains the corrected OpenDSS adapter, data manifests, chronological split, experiment scripts, tests, generated figures, and the manuscript source.

This manuscript is original, is not under consideration elsewhere, and has been approved by both authors. Siqiang Zhao led conceptualization, methodology, software, validation, analysis, investigation, data curation, visualization, and the original draft. Fengxiang Zhang supervised the study and reviewed and edited the manuscript. The authors declare no competing interests and received no dedicated external funding. OpenAI Codex/ChatGPT and Manus were used for code organization, literature checking, and language editing; the authors verified all content and remain responsible for the work.

Sincerely,
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